

Use-Case Flow Specification

Problem Statement #47 — Domain & SSL Certificate Expiry Alert System

Use Case ID / Name	UC-01: Monitor SSL Certificate Expiry
Primary Actor	SysAdmin (configures/receives alerts)
Supporting Actors	WHOIS/TLS Endpoint (external), Notification Gateway (external), Security Officer (escalation recipient)
Related Requirements	FR-001 (primary), FR-003, FR-004, NFR-001, NFR-002
Includes / Extends	«include» UC-04 Send Alert Notification «extend» UC-05 Escalate Unacknowledged Alert (extends UC-04)

Preconditions

- The list of monitored domains/certificates is configured and stored in the system database (see UC-07).
- The daily monitoring scheduler (cron job) is active and has outbound network access to each target domain on port 443.
- Alert thresholds (30 / 15 / 3 days) are configured, either at default or customized values (see UC-03).
- The Notification Gateway (email/SMS/Slack) is configured and reachable.

Postconditions

Success: All monitored certificates are scanned; expiration dates are recorded; alerts are queued/sent for any certificate within a configured threshold window; the scan result is written to the audit log.

Failure: Unreachable domains are logged with a retry count and flagged as "Unreachable" for SysAdmin review; the scan continues for remaining domains.

Main Success Scenario

1. The Scheduler triggers the daily SSL Certificate Monitoring job at the configured time (e.g., 02:00 UTC).
2. The system retrieves the current list of monitored domains/certificates from the database.
3. For each domain, the system opens a TLS handshake to the domain's HTTPS endpoint via the WHOIS/TLS Endpoint interface.
4. The system extracts the certificate's expiration date from the handshake response.
5. The system calculates the number of days remaining until expiry.
6. The system compares the remaining days against the configured thresholds (30, 15, 3 days).
7. If a threshold is met, the system executes the «include» use case Send Alert Notification (UC-04), passing the domain name, expiry date, and days remaining.
8. The Notification Gateway delivers the alert to the SysAdmin (email/SMS/Slack).
9. The system logs the scan result (domain, status, expiry date, timestamp) to the audit log.
10. The use case ends successfully once all monitored domains have been scanned.

Alternate Flows

3a. TLS Handshake Failure / Domain Unreachable

- 3a1. The system logs a "handshake failed" status for the domain, with timestamp and error code.
- 3a2. The system retries the handshake up to 2 additional times using exponential backoff (per NFR-002).
- 3a3. If all retries fail, the system marks the domain "Unreachable" and triggers Send Alert Notification (UC-04) to notify the SysAdmin of the monitoring failure (distinct from an expiry alert).
- 3a4. The use case resumes at step 2 for the next domain in the list.

8a. Alert Not Acknowledged (Escalation)

- 8a1. If the SysAdmin does not acknowledge the alert within the configured window (default 24 hours, per FR-004), the «extend» use case Escalate Unacknowledged Alert (UC-05) is triggered.
- 8a2. The alert severity is raised and re-sent to the Security Officer via Send Alert Notification (UC-04).
- 8a3. The escalation is recorded in the audit log with the original and escalated timestamps.